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Sub: OSSP practical 5b

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

1.

```
hasini@DESKTOP-UTCBQ51: ~$ cd ~/OSSP
hasini@DESKTOP-UTCBQ51:~/OSSP$ nano pipe.c
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];
    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

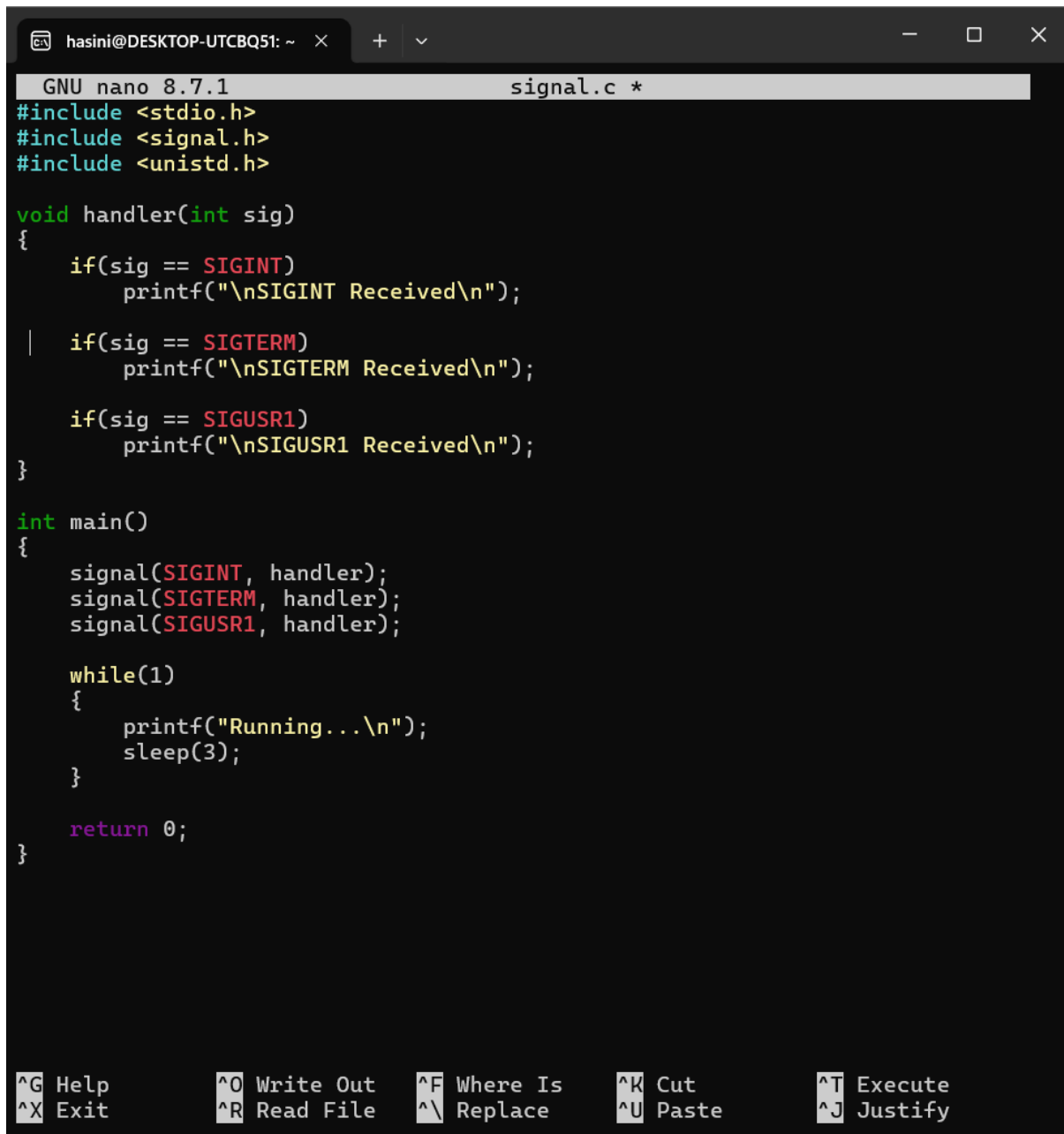
    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
hasini@DESKTOP-UTCBQ51:~/OSSP$ gcc pipe.c -o pipe
hasini@DESKTOP-UTCBQ51:~/OSSP$ ./pipe
drwxr-xr-x 2 hasini hasini 4096 Sep 18 06:39 include
-rw-r--r-- 1 hasini hasini 503 Sep 29 15:18 pipe.c
drwxr-xr-x 2 hasini hasini 4096 Sep 29 03:31 src
hasini@DESKTOP-UTCBQ51:~/OSSP$ |
```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.



The screenshot shows a terminal window with a dark background. At the top, a window title bar reads 'hasini@DESKTOP-UTCBO51: ~'. Below it, the terminal header shows 'GNU nano 8.7.1' and 'signal.c *'. The code is written in C and uses syntax highlighting: keywords are green, comments are grey, strings are yellow, and identifiers are white. The code defines a signal handler function 'handler' that prints messages for SIGINT, SIGTERM, and SIGUSR1. The 'main' function registers these signals and enters a loop that prints 'Running...\n' and sleeps for 3 seconds. At the bottom, a row of keyboard shortcuts is displayed: ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^F Where Is, ^\ Replace, ^K Cut, ^U Paste, ^T Execute, and ^J Justify.

```
GNU nano 8.7.1 signal.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig == SIGINT)
        printf("\nSIGINT Received\n");

    if(sig == SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig == SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

hasini@DESKTOP-UTCBO51: ~ X + v

```
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP
```

```
hasini@DESKTOP-UTCBO51:~/OSSP$ gcc signal.c -o signal
```

```
hasini@DESKTOP-UTCBO51:~/OSSP$ ./signal
```

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Running...
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Running...
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Running...
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Running...
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```
SIGUSR1 Received
```

```
Running...
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Running...
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Running...
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Running...
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Running...
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```
kill -SIGUSR1 530Running...
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hasini@DESKTOP-UTCQ51: ~

+

~

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Running...

Running...

Running...

SIGINT Received

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SIGINT Received

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Running...

Running...

Running...

Running...

Running...

Running...

Running...

Running...

Running...

Running...

Running...

Running...

[1]+ Stopped

./signal

hasini@DESKTOP-UTCQ51:~/OSSP\$ |



hasini@DESKTOP-UTCBQ51: ~



```
hasini@DESKTOP-UTCBQ51:~$ pidof signal
```

```
530
```

```
hasini@DESKTOP-UTCBQ51:~$ kill -SIGUSR1 530
```

```
hasini@DESKTOP-UTCBQ51:~$ kill -SIGTERM 530
```

```
hasini@DESKTOP-UTCBQ51:~$ kill -SIGINT 530
```

```
hasini@DESKTOP-UTCBQ51:~$
```